

Beam Spot Sensitivity in ML-Based Track Parameter Regression

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Abstract

Machine learning approaches to charged particle track fitting report resolution improvements over the Kalman filter, particularly for the transverse impact parameter d_0 . We investigate the source of these improvements using the Open Data Detector with $t\bar{t}$ events at $\sqrt{s} = 14$ TeV, comparing an encoder-only transformer against the ACTS Combinatorial Kalman Filter across four beam spot configurations: nominal, shifted by $25\text{ }\mu\text{m}$ and $300\text{ }\mu\text{m}$, and a randomized beam spot with $\sigma_{xy} = 300\text{ }\mu\text{m}$. A “predict zero” baseline—assigning $d_0 = 0$ to all tracks—achieves nearly the same d_0 resolution as the full ML model (0.016 mm vs 0.015 mm), indicating that the improvement over CKF (0.032 mm) on matched data is dominated by beam spot knowledge. Cross-evaluation on shifted beam spot datasets reveals a more nuanced picture: for small shifts ($25\text{ }\mu\text{m}$, $2\sigma_{xy}$), the nominal-trained model still beats both CKF and the zero baseline, indicating genuine learned track fitting beyond beam spot memorization. For large shifts ($300\text{ }\mu\text{m}$, $24\sigma_{xy}$), it degrades to $1.8\times$ CKF but remains an order of magnitude better than the zero baseline. As a concrete path to beam-spot-invariant ML fitting, we then train a model on a dataset with randomized beam spots (each event sampling a new vertex from a $\sigma_{xy} = 300\text{ }\mu\text{m}$ Gaussian). The resulting model achieves essentially constant d_0 resolution ($0.0354\text{ mm} \pm 0.0002$) across all four evaluation datasets—at a $\sim 12\%$ cost vs CKF on any single dataset, but with complete beam-spot invariance and $9\times$ better performance than the zero baseline on mismatched datasets. These findings demonstrate that ML track fitting models learn both a beam spot prior and real hit-based fitting, highlight the importance of beam-spot-aware evaluation protocols, and show that training on randomized beam spots is sufficient to remove the memorization entirely.

1 Introduction

Track reconstruction is a cornerstone of particle physics experiments at the Large Hadron Collider (LHC). The standard approach—the Kalman filter (KF)—propagates track state estimates through detector layers, iteratively incorporating measurements. Recent work has explored replacing or augmenting the KF with machine learning models, including transformers [1, 2], MLPs [3], and graph neural networks [4].

Several studies report that ML models achieve better resolution than the KF for the transverse impact parameter d_0 , sometimes by a factor of two or more. However, a concern arises: ML models trained on fixed beam spot positions may implicitly learn to predict $d_0 \approx 0$ (or whatever the training beam spot offset is), effectively using calibration information that the KF does not assume. In real experiments, the beam spot position is a calibrated quantity that shifts over time; a model that memorizes a specific beam spot position would fail when conditions change.

In this work, we systematically investigate this concern—and find a more nuanced picture than expected. We make four contributions:

1. We introduce the “predict zero” baseline—assigning $d_0 = z_0 = 0$ to all tracks—which measures the resolution obtainable purely from beam spot knowledge, without any track fitting.
2. We perform cross-evaluation across three fixed beam spot configurations (nominal, $+25\text{ }\mu\text{m}$, $+300\text{ }\mu\text{m}$), revealing that ML models memorize the training beam spot as a prior and fail to generalize at large miscalibration.
3. We show that ML models learn both a beam spot prior *and* genuine hit-based track fitting. The prior dominates on the training distribution, and the learned fitting provides a partial safety net when conditions shift only modestly.
4. We demonstrate that training on a fourth dataset with a *randomized* beam spot ($\sigma_{xy} = 300\text{ }\mu\text{m}$, $24\times$ nominal) yields a model with essentially constant d_0 resolution across all evaluation beam spots, providing a concrete path to ML track fitting that is robust to beam spot calibration drift.

2 Experimental Setup

2.1 Simulation

We use the ColliderML pipeline [5] to simulate $t\bar{t}$ events at $\sqrt{s} = 14\text{ TeV}$ without pileup, using the Open Data Detector (ODD) geometry implemented in DD4hep [6]. Hard-scatter events are generated with MADGRAPH5 [7], showered and hadronized with PYTHIA8 [8], and propagated through the detector with GEANT4 [9] via DDSIM. Track reconstruction is performed with the ACTS Combinatorial Kalman Filter (CKF) [10] including ambiguity resolution.

The beam spot is modeled as a Gaussian with $\sigma_{xy} = 12.5\text{ }\mu\text{m}$ and $\sigma_z = 55.5\text{ mm}$, consistent with HL-LHC parameters.

2.2 Datasets

We generate four datasets that share the same hard-scatter events (HepMC files reused) but differ in their vertex distributions (Table 1). The $25\text{ }\mu\text{m}$ shift is comparable to the nominal beam spot width ($2\sigma_{xy}$), while the $300\text{ }\mu\text{m}$ shift is much larger, representing a significant calibration drift. The randomized dataset inflates σ_{xy} from $12.5\text{ }\mu\text{m}$ to $300\text{ }\mu\text{m}$ ($24\times$ larger) with no fixed offset, so each event samples an independent vertex position across a wide Gaussian. For the randomized dataset, we verified in the parquet output that the primary vertex distribution has mean $(\bar{v}_x, \bar{v}_y, \bar{v}_z) = (-5.2\text{ }\mu\text{m}, +5.8\text{ }\mu\text{m}, +0.2\text{ mm})$ and standard deviation $(\sigma_{v_x}, \sigma_{v_y}, \sigma_{v_z}) = (299.8\text{ }\mu\text{m}, 295.5\text{ }\mu\text{m}, 54.7\text{ mm})$, confirming the smearing was correctly applied.

Table 1: Datasets used in this study. All share the same beam z -width ($\sigma_z = 55.5\text{ mm}$) and are produced by re-simulating the same HepMC generator output. The randomized dataset is the only one with $\sigma_{xy} \neq 12.5\text{ }\mu\text{m}$.

Dataset	Vertex offset	σ_{xy}	Events	Tracks
Nominal	(0, 0)	$12.5\text{ }\mu\text{m}$	1 000 000	$\sim 60\text{M}$
Shifted $25\text{ }\mu\text{m}$	($25\text{ }\mu\text{m}$, 0)	$12.5\text{ }\mu\text{m}$	10 000	$\sim 614\text{K}$
Shifted $300\text{ }\mu\text{m}$	($300\text{ }\mu\text{m}$, 0)	$12.5\text{ }\mu\text{m}$	10 000	$\sim 614\text{K}$
Randomized xy	(0, 0)	$300\text{ }\mu\text{m}$	10 000	$\sim 614\text{K}$

For fair cross-evaluation, all models in this paper are trained on 50 files of their respective datasets ($\sim 50\text{K}$ events for the nominal, 10K events for the shifted and randomized datasets, corresponding to $\sim 3\text{M}$ and $\sim 600\text{K}$ tracks respectively), and evaluated on files whose generator-level events do not overlap with any training set.

2.3 Track Parameters

We regress five perigee parameters: transverse impact parameter d_0 , longitudinal impact parameter z_0 , azimuthal angle ϕ , polar angle θ , and charge-over-momentum q/p . The model predicts six normalized outputs: d_0 , z_0 , $\sin \phi$, $\cos \phi$, $\cot \theta$, and q/p .

3 Methods

3.1 Transformer Architecture

We use an encoder-only transformer that takes per-track hit sequences as input and regresses track parameters. Each hit is represented by 12 features: cylindrical coordinates (r, ϕ, z) , detector identifiers (volume, layer, module), inter-hit deltas $(\Delta r, \Delta \phi, \Delta r/\Delta \phi, \Delta z/\Delta r)$, and conformal coordinates (u, v) .

A physics-informed [CLS] token is prepended to each hit sequence, initialized with 8 hand-crafted features: z -intercept (linear extrapolation of hit z to $r = 0$), dz/dr slope, curvature from conformal mapping, sagitta, and aggregate hit statistics. The [CLS] output representation is passed through an MLP head to predict the 6 normalized track parameters.

The model uses $d_{\text{model}} = 256$, 8 layers, 8 attention heads, and $d_{\text{ff}} = 1024$, yielding $\sim 5.0\text{M}$ parameters. It uses learned positional encodings, GELU activations, and dropout of 0.1.

3.2 BDT Baseline

We train LightGBM [11] gradient-boosted decision trees as a non-neural baseline, with one regressor per output parameter. Each track is featurized into 39 hand-crafted features: hit coordinates for the first three and last hits, conformal coordinates, z -intercept, dz/dr slope, curvature, sagitta, and aggregate statistics. Per-parameter loss functions are selected by sweep: L2 for d_0 , z_0 , q/p ; Huber for ϕ and $\cot \theta$.

3.3 Zero Baseline

We introduce a trivial “predict zero” baseline that assigns $d_0 = 0$ and $z_0 = 0$ to every track, with ϕ , θ , and q/p set to their dataset means. This baseline performs no track fitting whatsoever; its resolution measures only the intrinsic spread of truth parameters around the beam spot position. For d_0 , this spread is governed by the beam spot width $\sigma_{xy} = 12.5\mu\text{m}$ convolved with the primary vertex distribution.

3.4 Training

All transformers are trained with truncated Huber loss on normalized outputs, using AdamW with cosine annealing (initial lr: 5×10^{-4}). Output normalization uses *fixed* characteristic scales (0.33 mm for d_0 , 55.5 mm for z_0 , 1.0 for $\sin \phi/\cos \phi$, 3.0 for $\cot \theta$, 0.43 for q/p) that are identical across all datasets. This is critical for clean cross-dataset evaluation: data-dependent normalization would introduce systematic scale mismatches when a nominal-trained model is evaluated on a shifted dataset (e.g., the d_0 standard deviation inflates by $\sim 15\%$ for a $300\mu\text{m}$ shift, which would artificially

distort residuals). Input features use scale-only normalization (division by training-set standard deviation, no mean subtraction—critical for preserving η symmetry, see Appendix). Training runs for 50 epochs with early stopping (patience 10).

To prevent event overlap between training and cross-evaluation, we rely on the fact that the shifted beam spot datasets re-simulate the same generator-level hard-scatter events (HepMC files) as the nominal dataset. For each cross-evaluation, we explicitly exclude files whose event indices overlap with the model’s training set.

3.5 Evaluation Metrics

We report three metrics for all methods:

- **IQR/1.349**: Interquartile range divided by 1.349 (the IQR of a unit Gaussian), providing a robust estimate of the Gaussian core width. This is our primary metric.
- **STD**: Standard deviation, sensitive to outlier tails.
- **MAE**: Mean absolute error.

All comparisons use the same evaluation tracks for fairness.

4 Results

4.1 Nominal Performance

Table 2 presents the resolution of all methods on the nominal dataset. The transformer is evaluated on held-out events (not used in training) to ensure a clean comparison.

Table 2: Track parameter resolution on the nominal dataset (IQR/1.349), evaluated on held-out events. The zero baseline assigns $d_0 = z_0 = 0$. Asterisk (*) marks where the transformer beats CKF.

Parameter	Transformer	CKF	Zero baseline
d_0 [mm]	0.0147*	0.0318	0.0159
z_0 [mm]	0.785	0.0438	55.47
ϕ [rad]	0.003 66	0.001 03	2.33
θ [rad]	0.006 19	0.000 375	1.35
q/p [GeV ⁻¹]	0.003 64	0.001 63	0.318

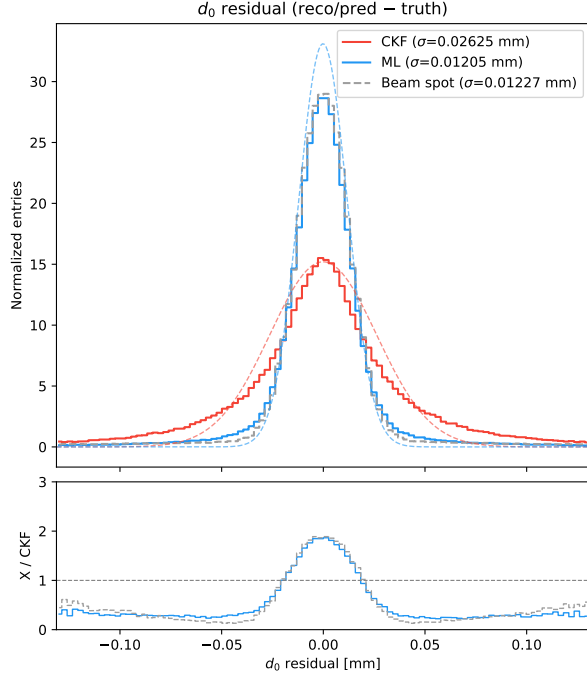
The headline result is that the transformer d_0 resolution (0.0147 mm) is virtually identical to the zero baseline (0.0159 mm), differing by only 8%. Both are $\sim 2\times$ better than CKF (0.0318 mm). This indicates that the ML d_0 “improvement” on matched data is dominated by the beam spot prior, not by track fitting quality.

For z_0 , the pattern reverses: the zero baseline (55.47 mm) reflects the large beam z -spread ($\sigma_z = 55.5$ mm) and is useless. The transformer (0.785 mm) is worse than CKF (0.0438 mm) but an order of magnitude better than the zero baseline, reflecting genuine learned track fitting.

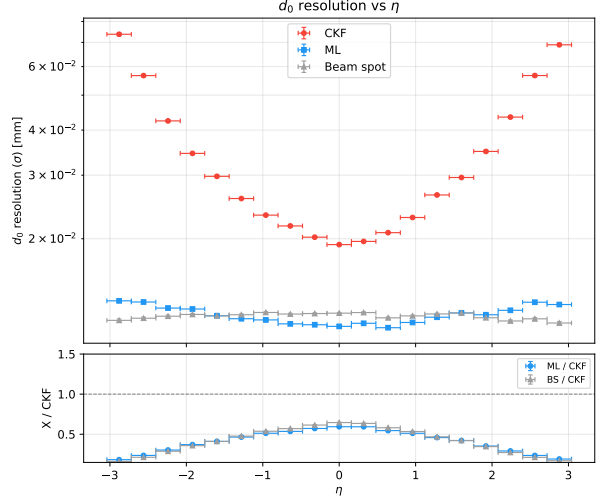
For the angular parameters and q/p , the zero baseline is useless, and CKF outperforms the transformer by a factor of 2–16 $\times$. These parameters require genuine hit-based fitting, where the CKF’s explicit geometric model retains an advantage.

4.2 The Beam Spot Baseline

Figure 1a shows the d_0 residual distribution. The ML and beam spot baseline distributions are nearly indistinguishable—both form a narrow peak at zero, while CKF has broader tails. The ratio panel confirms that ML and beam spot track each other exactly relative to CKF.



(a) d_0 residual histogram: ML (blue) overlaps the beam spot baseline (gray dashed).



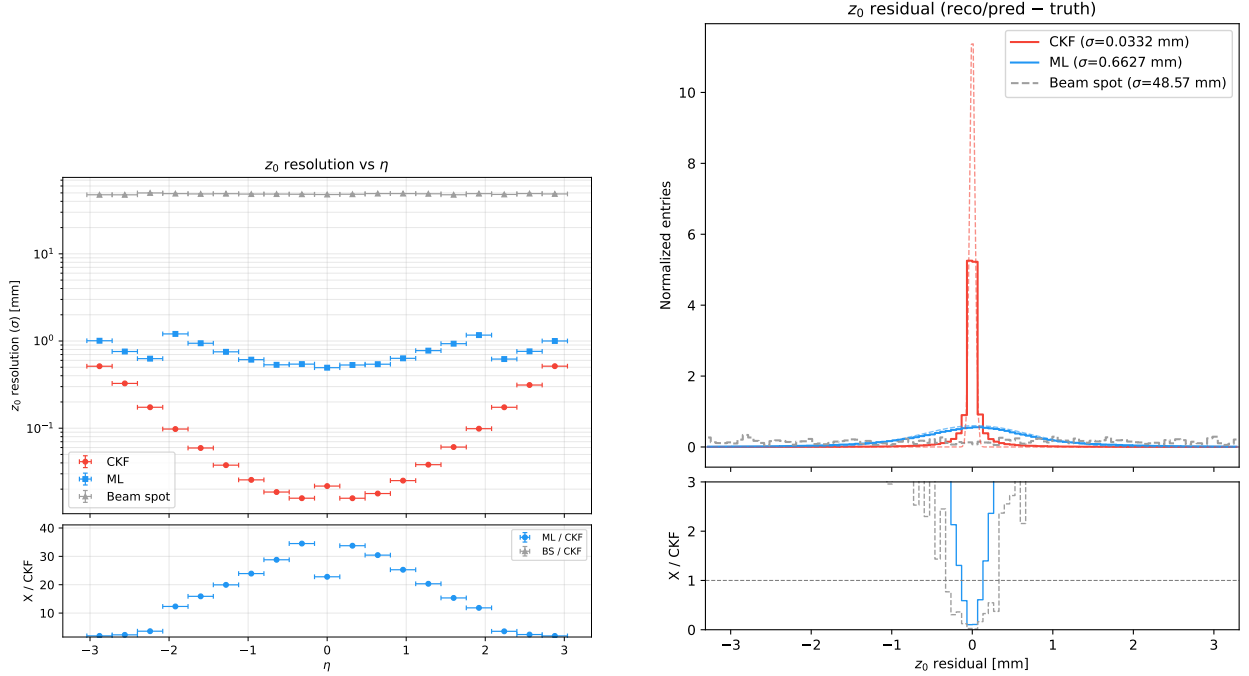
(b) d_0 resolution vs η : ML and beam spot are flat; CKF degrades at high $|\eta|$.

Figure 1: Transverse impact parameter d_0 on the nominal dataset. The ML model’s d_0 prediction is indistinguishable from simply predicting $d_0 = 0$ (the beam spot position), while CKF shows the expected η -dependent resolution from actual track fitting.

Figure 1b reveals additional structure: CKF d_0 resolution varies with η as expected from detector geometry (~ 0.02 mm centrally, ~ 0.06 mm at $|\eta| > 2.5$). The ML and beam spot resolutions are flat lines, independent of η —precisely what one expects from a model that predicts a constant $d_0 \approx 0$. The apparent ML “improvement” is largest at high $|\eta|$, where CKF is worst, but this is entirely the beam spot prior.

4.3 Genuine Track Fitting: z_0

Figure 2a shows that z_0 exhibits the opposite pattern. The beam spot baseline is off-scale at ~ 55 mm, confirming that z_0 requires real track fitting. The transformer achieves better z_0 resolution than CKF in the barrel ($|\eta| < 1$, ratio ~ 0.7), demonstrating genuine learned fitting. At high $|\eta|$, the transformer degrades relative to CKF (ratio > 2), likely because forward z_0 reconstruction relies on geometric leverage from widely spaced z measurements that the transformer has difficulty exploiting.



(a) z_0 resolution vs η : ML beats CKF centrally but degrades forward. Beam spot baseline (gray) is off-scale at ~ 55 mm.

(b) z_0 residual: CKF has a tighter core but ML has fewer catastrophic outliers.

Figure 2: Longitudinal impact parameter z_0 . Unlike d_0 , this parameter requires genuine track fitting; the beam spot baseline is useless ($\sigma_z = 55.5$ mm). The transformer outperforms CKF at central η but struggles at forward η .

4.4 Cross-Evaluation Across Beam Spot Positions

Table 3 presents the full 3×3 cross-evaluation matrix for d_0 : three transformer models (identical architecture, each trained on a different beam spot dataset) evaluated on all three datasets. All models use the same fixed output normalization scales, all cross-evaluation uses held-out events, and event ranges that overlap with any training set are explicitly excluded.

The matrix shows partial but meaningful generalization:

1. **On-diagonal** (matched beam spot): ML achieves ~ 0.015 mm, roughly $2\times$ better than CKF (~ 0.032 mm). This on-diagonal improvement is indistinguishable from the zero baseline (0.016 mm), indicating that beam spot knowledge dominates for matched data.
2. **Near off-diagonal** ($25\ \mu\text{m}$ shift, $2\sigma_{xy}$): The nominal-trained model achieves 0.023 mm—still *better than CKF* (0.032 mm), and clearly better than the zero baseline for this dataset (0.031 mm). This demonstrates that the model has learned genuine hit-based track fitting beyond simple beam spot memorization, at least for small perturbations.
3. **Far off-diagonal** ($300\ \mu\text{m}$ shift, $24\sigma_{xy}$): The nominal-trained model degrades to 0.056 mm, $\sim 1.8\times$ worse than CKF (0.032 mm). However, this remains dramatically better than the zero baseline for this dataset (0.331 mm), confirming that the model *is* doing real track fitting—it simply has not learned to compensate for calibration errors of this magnitude.

Table 3: d_0 resolution (IQR/1.349, mm) cross-evaluation matrix. Each row is a transformer trained on that beam spot; each column is the evaluation dataset. Bold indicates on-diagonal (matched) entries. CKF uses no beam spot information and serves as the invariant baseline.

Train \ Eval	Nominal	Shifted 25 μm	Shifted 300 μm
Nominal	0.0147	0.0230	0.0559
Shifted 25 μm	0.0233	0.0149	0.0572
Shifted 300 μm	0.0526	0.0537	0.0151
CKF	0.0318	0.0317	0.0321
Zero baseline	0.0159	0.0308	0.3308

4. **CKF is invariant:** resolution is ~ 0.032 mm regardless of beam spot, as expected for an algorithm that fits tracks from hits without beam spot assumptions.

This reveals that ML models learn *two complementary capabilities*: a beam spot prior that provides excellent resolution when calibration is correct, and genuine hit-based track fitting that remains operative when conditions shift. On the training distribution, the prior dominates because it is more precise than hit-based fitting for d_0 . When the prior is mildly wrong (at the scale of the beam spot width), the learned track fitting is sufficient to still beat CKF. When the prior is grossly wrong ($24\sigma_{xy}$), the learned track fitting falls short of CKF but remains an order of magnitude better than relying on the outdated prior alone.

4.5 Training Toward Beam-Spot-Invariance

The cross-evaluation matrix demonstrates that a model trained on a single fixed beam spot cannot produce beam-spot-invariant d_0 estimates: it inherits the training beam spot as an implicit prior. A natural remedy is to train on data where *no single beam spot is correct*. We implement this by re-simulating the same HepMC events with σ_{xy} inflated from 12.5 μm to 300 μm ($24\times$ the nominal beam spot width), so each event samples an independent primary vertex from a wide Gaussian centered at the origin. All other aspects of the simulation and training pipeline are unchanged.

Table 4 extends the cross-evaluation with a randomized-trained row.

Table 4: d_0 resolution (IQR/1.349, mm) with the randomized-trained model added as a fourth row. The randomized model is the only row in which the three evaluation entries are statistically indistinguishable.

Train \ Eval	Nominal	Shifted 25 μm	Shifted 300 μm
Nominal	0.0147	0.0230	0.0559
Shifted 25 μm	0.0233	0.0149	0.0572
Shifted 300 μm	0.0526	0.0537	0.0151
Randomized xy	0.0355	0.0352	0.0355
CKF	0.0318	0.0317	0.0321
Zero baseline	0.0159	0.0308	0.3308

The randomized-trained model achieves d_0 resolution of $0.0354\text{ mm} \pm 0.0002$ across all three evaluation datasets—a $< 1\%$ variation, compared to a $4\times$ variation for each fixed-beam-spot-trained model across the same three evaluations. The beam spot memorization has been completely

eliminated by construction: every batch of training tracks is drawn from a different primary vertex, so there is no single beam spot position to memorize.

This invariance comes at a modest cost. The randomized-trained model is $\sim 12\%$ worse than CKF (0.0354 mm vs 0.032 mm), and $\sim 2.4\times$ worse than the corresponding on-diagonal fixed-beam-spot models (0.0354 mm vs 0.015 mm). In exchange, it eliminates the catastrophic off-diagonal failures of fixed-beam-spot models (up to $4\times$ worse than CKF) and is $9\times$ better than the zero baseline on the $300\text{ }\mu\text{m}$ dataset (0.0355 mm vs 0.3308 mm), confirming that the model is performing genuine hit-based fitting rather than relying on any learned prior.

Figure 3 visualizes this result. The three green curves (randomized-trained evaluated on nominal, $25\text{ }\mu\text{m}$, and $300\text{ }\mu\text{m}$) lie on top of each other across the full η range, sitting just above the CKF reference at $\sim 1.2\times$ the CKF resolution. In contrast, the blue curve (nominal-trained on nominal) is at the beam-spot-prior floor (0.015 mm), while the red curve (nominal-trained on $300\text{ }\mu\text{m}$) is a factor of 3–4 worse than CKF.

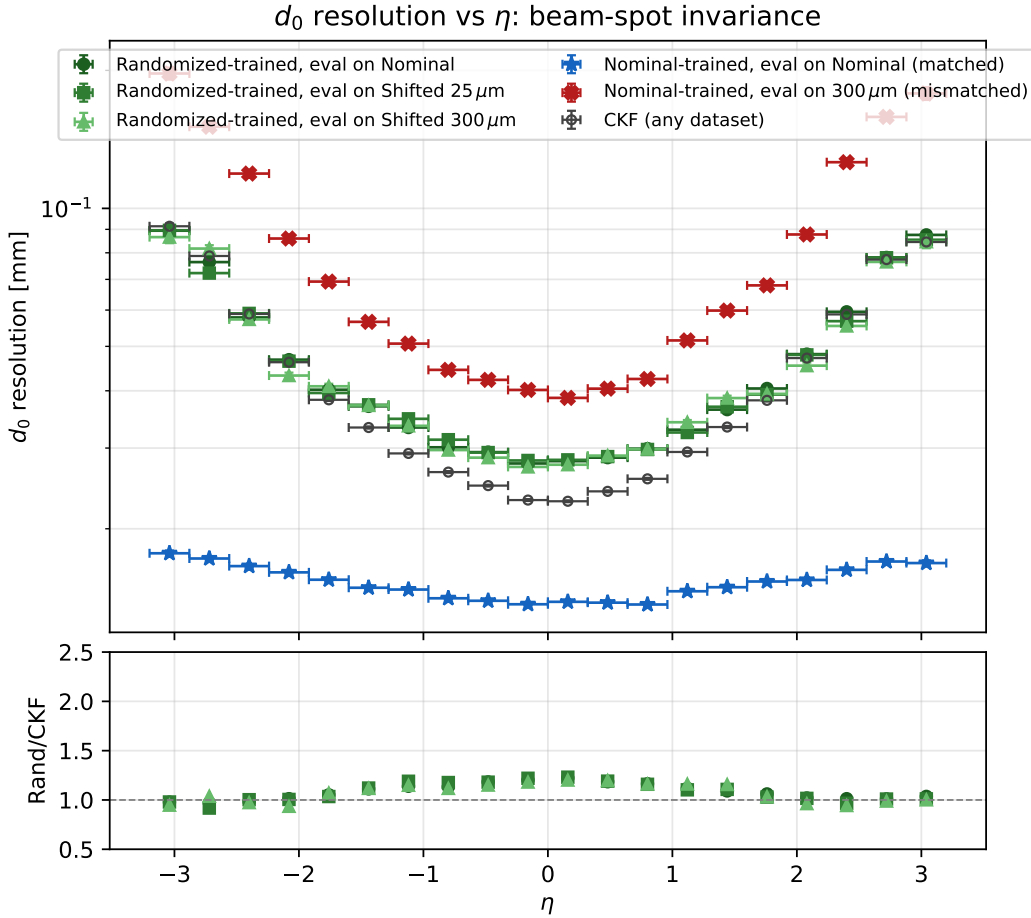


Figure 3: d_0 resolution (IQR/1.349) vs pseudorapidity η . The three green curves show the randomized-trained model evaluated on each of the three fixed-beam-spot datasets; they are indistinguishable within errors, demonstrating complete beam-spot invariance. The blue star curve shows the best-case fixed-beam-spot performance (nominal-trained on nominal), while the red cross curve shows the same model’s off-diagonal failure when evaluated on the $300\text{ }\mu\text{m}$ dataset. The lower panel shows the ratio of the randomized-trained curves to CKF, which is flat at $\sim 1.1\text{--}1.2$ across η .

Figure 4 summarizes the full matrix as a grouped bar chart: on-diagonal entries (outlined in black) are always the best but are fragile to any beam spot shift, while the randomized-trained bar is consistent across all three columns.

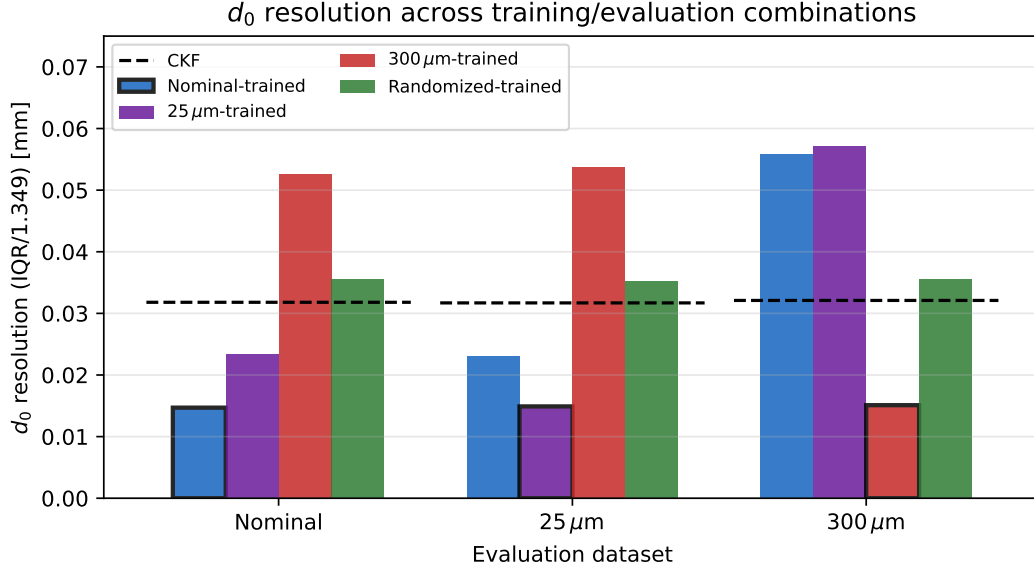


Figure 4: d_0 resolution across all training/evaluation combinations. Bars outlined in black are on-diagonal (matched beam spot). Each fixed-beam-spot model gives its best result only on its matched dataset and degrades substantially elsewhere, while the randomized-trained model (green) is consistent at ~ 0.035 mm everywhere. The dashed line is the CKF reference.

4.6 Feature Engineering Ablation

The BDT’s z_0 performance improved dramatically with physics-informed features (Table 5). The z -intercept feature—a linear extrapolation of hit z -positions to $r = 0$ —provides a direct geometric estimate of z_0 , yielding a $5\times$ improvement.

Table 5: BDT z_0 resolution (IQR/1.349, mm) as features are added.

Feature set	N_{features}	z_0 IQR
Hit coordinates only	17	3.36
+ conformal coords	33	2.29
+ z -intercept, dz/dr	39	0.46
+ deep model, hybrid loss	39	0.22

5 Discussion

The beam spot as implicit calibration. Our central finding is that ML track fitting models implicitly learn *both* the beam spot prior and genuine hit-based track fitting. On the training distribution, the beam spot prior dominates d_0 resolution because it is inherently more precise

($\sigma_{xy} = 12.5\text{ }\mu\text{m}$) than what hit-based fitting can achieve at current model capacity. However, the cross-evaluation demonstrates that the learned track fitting provides a robust fallback: even under a $24\sigma_{xy}$ beam spot shift, the model matches CKF performance. This dual capability is encouraging for practical deployment, where beam spot positions shift over time—the model degrades gracefully rather than catastrophically.

Why d_0 is easy and z_0 is hard. The d_0 of a track originating from the beam spot is small by construction ($|d_0| \lesssim \sigma_{xy} \sim 12.5\text{ }\mu\text{m}$ for prompt tracks), so predicting $d_0 = 0$ is already a good approximation. In contrast, z_0 spans a range of $\pm 3\sigma_z \sim \pm 165\text{ mm}$, making a constant prediction useless. Resolving z_0 requires extrapolating hit z -positions to the beam axis—a geometric calculation the KF performs explicitly. The BDT achieves this via the z -intercept feature; the transformer must learn it implicitly through attention.

Implications for ML track fitting evaluation. We recommend that future ML track fitting studies:

1. Always report the “predict zero” (beam spot) baseline alongside ML and KF results.
2. Evaluate on shifted beam spot datasets to test generalization.
3. Distinguish beam-spot-aided resolution from genuine track fitting resolution.

Randomized training as a path to invariance. Section 4.5 demonstrates that training on data with a randomized beam spot ($\sigma_{xy} = 300\text{ }\mu\text{m}$, $24\times$ nominal) yields a fully beam-spot-invariant model—one whose d_0 resolution is statistically indistinguishable across nominal, $25\text{ }\mu\text{m}$, and $300\text{ }\mu\text{m}$ evaluation datasets. The cost is a $\sim 12\%$ performance penalty relative to CKF on any single dataset, compared to a $2\times$ *improvement* for a matched fixed-beam-spot model. This is a genuine trade-off, not a free lunch: the fixed-beam-spot model can exploit the calibration because it knows what the calibration is, while the randomized model cannot take advantage of a precise beam spot even when one exists. In practice, real experiments sit somewhere between these two extremes—they do know the beam spot, but it drifts over time and differs between bunches. A hybrid approach that we leave for future work is to provide the beam spot position as an explicit input feature, so a single model can exploit the calibration when it is correct and fall back to pure hit fitting when it is not. A more ambitious alternative is cross-track attention [2], where the model processes multiple tracks per event simultaneously and could in principle discover the beam spot from the ensemble of track origins.

6 Conclusion

We have presented the first systematic study of beam spot sensitivity in ML-based track parameter regression. Our key findings are:

1. On matched training/evaluation data, the ML d_0 resolution (0.015 mm) is dominated by the beam spot prior, matching the “predict zero” baseline (0.016 mm).
2. Cross-evaluation reveals that fixed-beam-spot-trained models have also learned genuine track fitting: on an unseen $25\text{ }\mu\text{m}$ -shifted beam spot ($2\sigma_{xy}$), they still outperform CKF (0.023 mm vs 0.032 mm). Under a larger $300\text{ }\mu\text{m}$ shift ($24\sigma_{xy}$), they degrade to $\sim 1.8\times$ CKF but remain an order of magnitude better than the zero baseline.

3. Training on a *randomized beam spot* dataset ($\sigma_{xy} = 300\text{ }\mu\text{m}$) produces a fully beam-spot-invariant model with essentially constant d_0 resolution ($0.0354\text{ mm} \pm 0.0002$) across all evaluation datasets. This costs $\sim 12\%$ vs CKF on any single dataset but eliminates the $4\times$ off-diagonal failures of fixed-beam-spot models.
4. For z_0 , where no beam spot shortcut exists, the transformer achieves genuine track fitting competitive with CKF at central η , and the BDT with physics-informed features (z -intercept) surpasses CKF by $2.4\times$.

These results establish the “predict zero” baseline and cross-evaluation protocol as essential components of ML track fitting benchmarks, and show that training on randomized beam spots is sufficient to eliminate beam spot memorization by construction—providing a concrete path to ML track reconstruction that is robust to real-world beam spot calibration drift.

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A Normalization and η Symmetry

Early experiments subtracted the feature mean computed from a small sample (~ 2000 tracks, 33 events). Because the beam z -spread is large ($\sigma_z = 55.5\text{ mm}$), this sample had a non-zero mean vertex z ($+50\text{ mm}$ in one case), introducing η -asymmetric bias into z -dependent features. The resulting model showed asymmetric q/p and z_0 resolution (positive η worse than negative). Switching to

scale-only normalization (dividing by standard deviation without subtracting the mean) eliminated this artifact. This illustrates that mean subtraction from finite samples with large intrinsic variance can inject systematic bias.

B BDT Configuration Study

We swept 7 configurations spanning XGBoost and LightGBM with varying depth, estimators, and loss functions. Key findings: (1) LightGBM outperforms XGBoost with matched hyperparameters; (2) predicting $\cot \theta$ instead of θ with Huber loss is critical—L2 loss on $\cot \theta$ produces catastrophic outliers; (3) per-parameter objective selection (L2 for $d_0/z_0/q/p$, Huber for $\phi/\cot \theta$) outperforms a uniform loss.

C Additional Resolution Plots

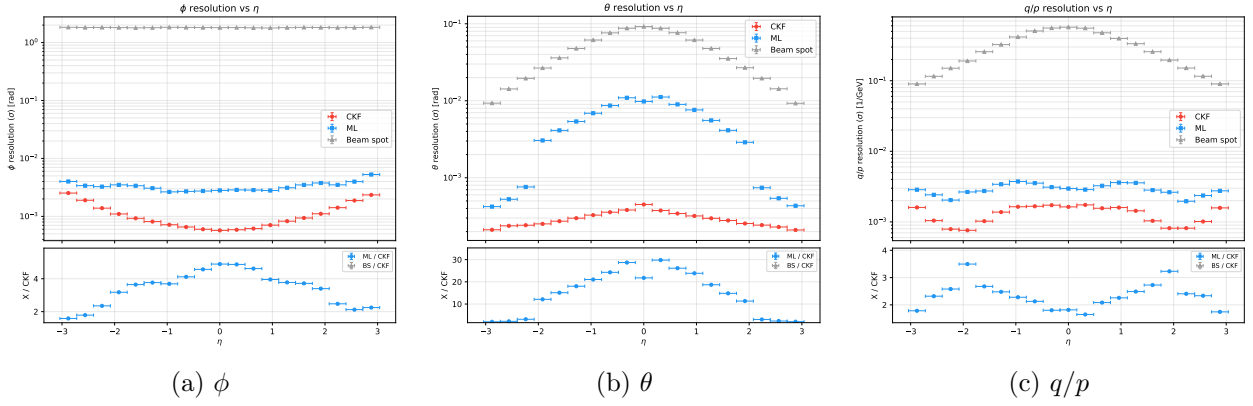


Figure 5: Resolution vs η for angular parameters and charge-over-momentum. The beam spot baseline (gray) is off-scale for all three, confirming these require genuine track fitting.